Accident in a Co-generation Plant

Analysis of the Human and Organisational Factors (HOF) involved

Minor Assignment | Technology Risk Governance

MSc in Cyber Risk Strategy and Governance | Prof. Paolo Trucco

Team members

X

Y

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Approach and reading key

Method

Each event of the accident time-line is analysed with the **behavioural approach** (Reason's model of organisational accidents), starting from the *phenotype* (how the event manifests) and going back to the *genotypes* (local conditions and latent factors). For every event we report:

- **one page** with the template compiled in bullet form, following the scheme used in class for Events 1 and 3: human factors pathway on the left, system factors pathway on the right, organisational latent errors on top;
- **one page** of justification, explaining which information of the case text supports each box, the classification of the active error and a short **cognitive view (CREAM)** of the human contribution.

Events 1 and 3 reproduce and complete the solution discussed in class; the other events are our own analysis.

How to read the templates

- **Event**: the event occurs if at least one of the two is present: a failure of a technical defence, barrier or safeguard, or an active human error.
- Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it, with the technical latent error behind it). **White boxes**: contributing conditions.
- **Top box vs. side boxes** (scope of influence): a latent factor that affects only one pathway goes into that pathway's box; a factor that affects both the human and the technical side goes on top.

Sources

[§x.y] section of the case document; [Fig. 3] event time-line; [Class] discussion of the case in class; [Inf.] our inference, not stated explicitly in the case.

Event time-line and overview

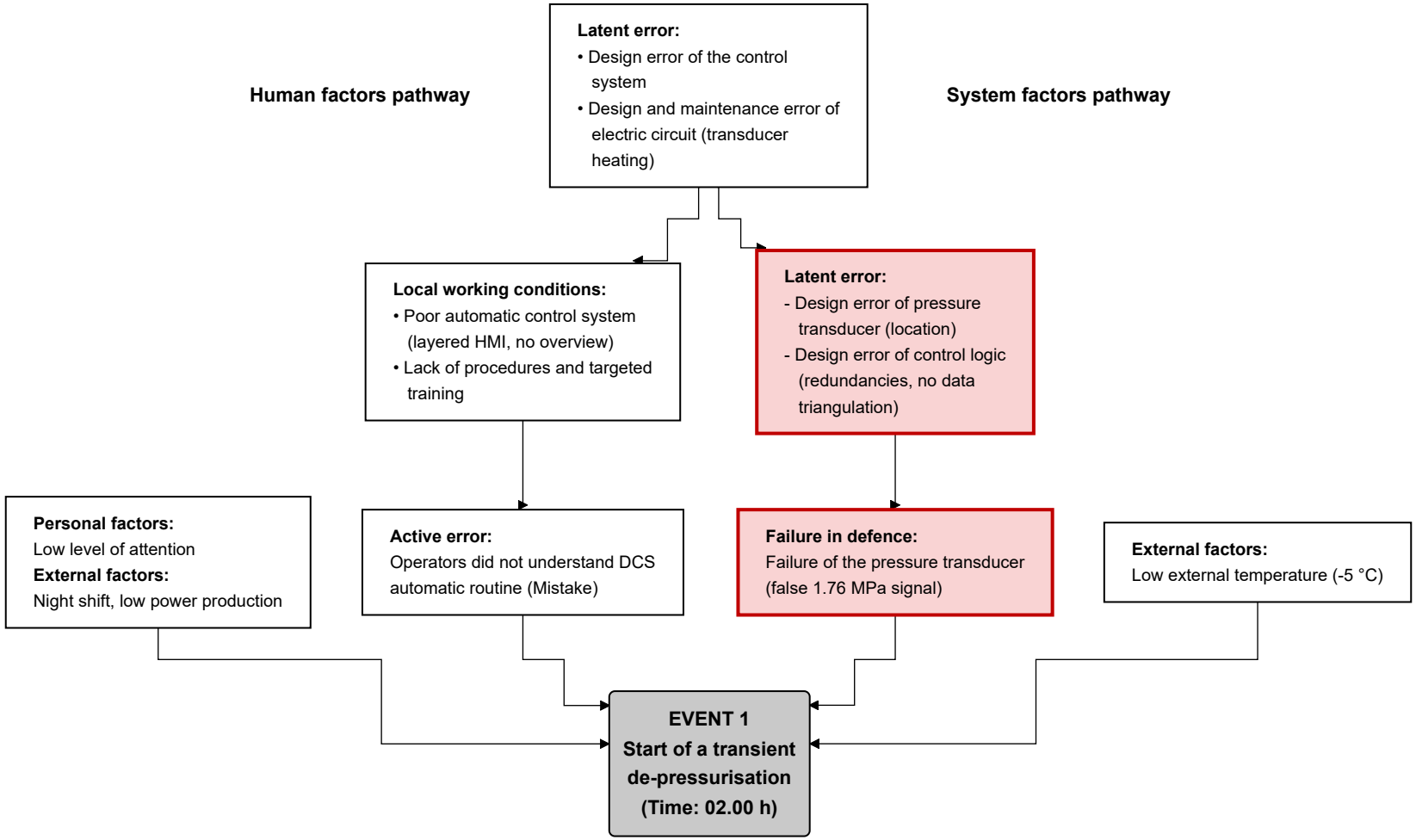
#	Time	Event	Dominant pathway
1	02.00	Pressure transducer failure, start of de-pressurisation	System
2	02.04	Manual activation of main circulation pumps	Human + system
3	02.41	Opening of relief valves, overflow of storage tank, flooding of pumping room	Human + system
4	03.00	Manual closure of relief valves (very late)	Human + system
5	03.20	Fire protection intervention, remote control out of service	System
6	03.45	Temperature increase at heat-exchanger outlet	Human + system
7	04.39	Circulation pumps started by-passing low-pressure protection	Human (violation) + system
8	05.15	Manual start-up of heat recovery from turbo-gas	Human + system
9	06.15	Shock wave (water hammer)	Final event

Emergency phases (causal and contributing factors)

- **From the control room**: 1st de-pressurisation and vaporisation (Events 1, 2, with 4).
- **On the field**, after the loss of remote control: 2nd de-pressurisation (3, 5), vaporisation due to heat exchange (4, 6, 8) and peak pressurisation (7), all converging on the shock wave (9).

Event 1 | Start of a transient de-pressurisation (02.00 h)

The outdoor pressure transducer freezes during a cold winter night and sends a false reading of 1.76 MPa while the real pressure is normal. The automatic control system reacts to the false signal: pressurising pumps slowed down, relief and expansion-tank valves opened, main feed pumps stopped. A rapid de-pressurisation starts.



Active error: Omission (Swain & Guttman) | Mistake (Reason) | Knowledge-based (SRK) | CREAM: Timing → Omission. **Dominant pathway**: system (technical failure of the pressure measurement).

Red boxes: direct causal chain of the event (active error and/or failure in defence that produced it). White boxes: contributing conditions. Top box: latent factors influencing both pathways. Template reproduces the class solution; items in parentheses complete it with the class discussion.

Event 1 | Justification of the template

Evidence from the case, box by box

Failure in defence (red). The transducer, installed outdoors near an expansion tank, froze together with its signal piping in a cold but "not exceptional" night and sent 1.76 MPa while the real pressure was normal [§2, ev. 1]. **External factors:** low external temperature, about -5 °C [Class].

Latent error, system (red). A safety-critical measurement was placed outdoors, protected only by an electric heating circuit (*location*) [§2; Class]. One single transducer drove both pump speed and pressure valves [§2]: no redundant source and no *data triangulation*, although 1.76 MPa was thermodynamically incompatible with the measured temperature and flow [Class]. The heater failure was also invisible, because the monitoring of the heaters' power supply was wrongly designed [§3.3].

The DCS then did exactly what it was designed for, but on wrong data: it acted as a barrier against a non-existent over-pressure. This is a loss of data *integrity*: the same scenario could be triggered by a cyber attack on the sensor signal [Class].

Active error (white). Operators did not understand what the automatic routine was doing and lost the chance to build situational awareness (*mistake*). Evidence comes from the phenotype of Event 2: their first action was to restart the circulation pumps [Class]. The box is white because the error did not create the de-pressurisation; it removed an early recovery opportunity.

Local working conditions. HMI made of layered windows, without an overview, and no warning for an "abnormal measure" out of the transducer range [§3.3]; no emergency procedures [§3.4]; training only on the job [§3.7].

Personal and external factors. Night shift with base-load demand: almost no interaction between operators and system, hence low attention and low readiness when the transient started [§3.6; Class].

Latent error, organisational (top). Design error of the control system and design & maintenance error of the electric circuit influence *both* pathways: they make the barrier fail and leave operators unable to understand it. Maintenance is also organisational: not providing procedures and training means not maintaining the operators' situational awareness as a defence [Class].

Classification of the active error

Phenotype (Swain & Guttman)	Omission: no diagnosis of, and no intervention on, the automatic routine.
Reason's taxonomy	Mistake: wrong understanding of the situation at the intention stage.
SRK level (Rasmussen)	Knowledge-based (diagnose a DCS acting on false data); the rule-based support was missing (no procedure for this transient).

Cognitive view (CREAM, Hollnagel 1998)

Socio-technical context	Organisation: no emergency procedures, shift team undersized for emergencies. Technology: layered HMI, no out-of-range alarm. Environment: night shift, stable base load.
Cognitive function	Observation $\rightarrow$ Interpretation (the automatic actions were not noticed or not interpreted).
Individual factors	Psychological aspects (low attention at night); experience and skill (on-the-job training only).
Error mode	Timing $\rightarrow$ Omission.
General antecedents	Inattention; Observation missed; Faulty diagnosis; Inadequate procedure.
Specific antecedent	None (first event of the sequence).

Improvement levers

- **Prevention**: relocated and redundant pressure sensors with plausibility checks (pressure vs. temperature and flow); alarm on heater supply failure.
- **Recovery**: out-of-range ("abnormal measure") alarm and an overview display of the automatic actions in progress.
- **Emergency management**: procedure and training for the de-pressurisation transient; integrity monitoring of OT sensor data (cyber).
- After the accident the company relocated and duplicated the sensors and replaced most of the DCS [Class].